

SimCSE :

Simple Contrastive Learning of Sentence Embedding

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Background Knowledge

- ✓ Contrastive Learning
- ✓ Positive instances
- ✓ Alignment and Uniformity

1. Background Knowledge

Contrastive Learning

$$\ell_i = -\log \frac{e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau}}{\sum_{j=1}^N e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)/\tau}},$$

☑ representation 이 가까울수록 Similarity ↑

Positive Instances

$$(x_i, x_i^+)$$

where x_i and x_i^+ are semanti- cally related.

☑ Unsupervised SimCSE를 learning 을 위해 positive pair 중요

Alignment & Uniformity

Alignment

$$\ell_{\text{align}} \triangleq \mathbb{E}_{(x, x^+) \sim p_{\text{pos}}} \|f(x) - f(x^+)\|^2,$$

☑ positive paired instance 사이의 거리

Uniformity

$$\ell_{\text{uniform}} \triangleq \log \mathbb{E}_{x, y \stackrel{i.i.d.}{\sim} p_{\text{data}}} e^{-2\|f(x) - f(y)\|^2}$$

☑ embedding이 uniform 하게 분포

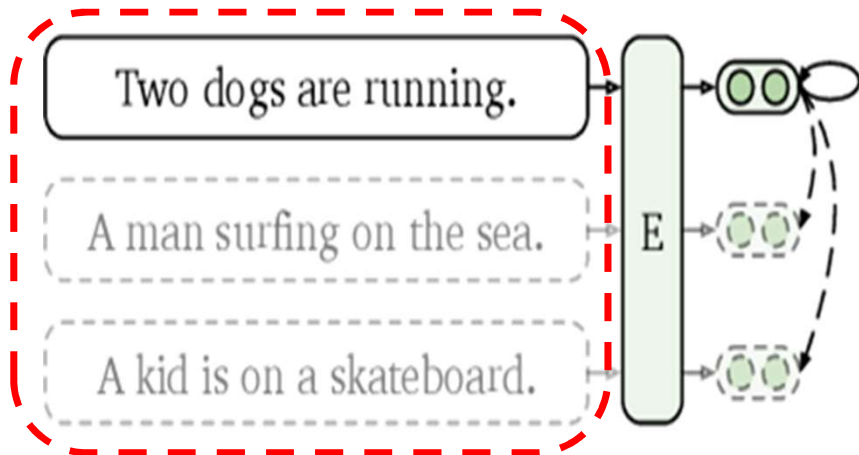
☑ Alignment 와 Uniformity 만족하게 학습

2 Introduction

- ✓ Contrastive Sentence Embedding
- ✓ Approach : Unsupervised / Supervised
- ✓ Sentence Representation 분석
 - Alignment
 - Uniformity

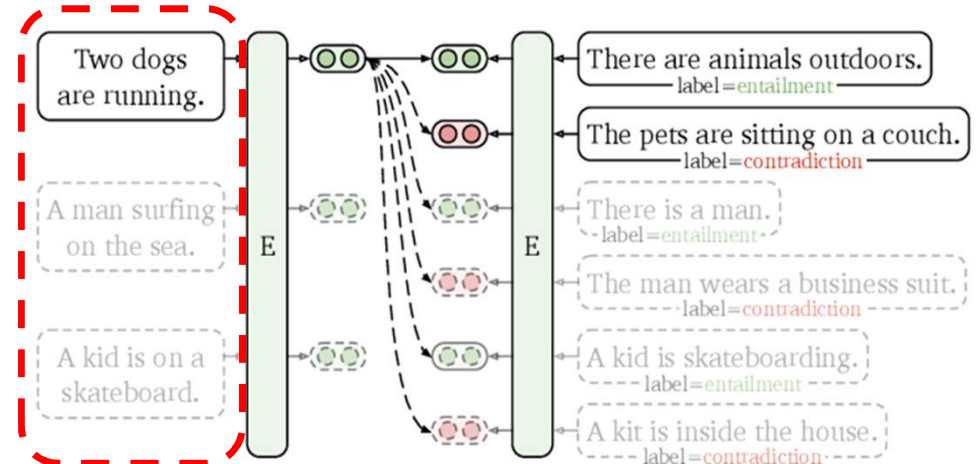
2. Introduction

Unsupervised SimCSE



- ☑ 동일한 문장을 input으로 넣어 Positive pair
- ☑ Dropout (noise) 활용한 minimal augmentation
- ☑ dropout 을 바꾸어 나온 hidden Representation을 positive pair 로 구성

Supervised SimCSE



- ☑ NLI(Natural Language Inference) 데이터 활용
- ☑ 'entailment' label을 가진 쌍의 문장을 positive pair
- ☑ 'contradiction' label을 가진 쌍의 문장을 Hard negative pair

3 SimCSE

- ✓ Unsupervised Approach
 - Dropout noise as augmentation
 - Alignment & Uniformity
- ✓ Supervised Approach
 - NLI data
 - Choice of labeled data

3. Unsupervised SimCSE

Dropout noise as augmentation

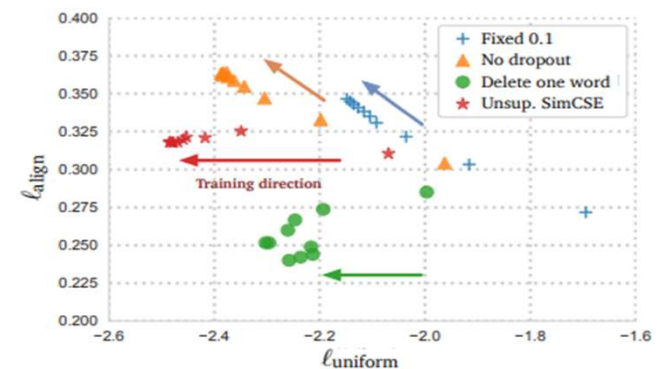
Data augmentation		STS-B	
None (unsup. SimCSE)		82.5	
Crop	10%	20%	30%
	77.8	71.4	63.6
Word deletion	10%	20%	30%
	75.9	72.2	68.2
Delete one word		75.9	
w/o dropout		74.2	
Synonym replacement		77.4	
MLM 15%		62.2	

Training objective	f_θ	$(f_{\theta_1}, f_{\theta_2})$
Next sentence	67.1	68.9
Next 3 sentences	67.4	68.8
Delete one word	75.9	73.1
Unsupervised SimCSE	82.5	80.7

- ☑ Dropout → minimal augmentation 효과
- ☑ Semantic Textual similarity 학습을 위한
NLI 데이터셋 구축, dual encoder 로 representation

Align & Uniform

p	0.0	0.01	0.05	0.1
STS-B	71.1	72.6	81.1	82.5
p	0.15	0.2	0.5	Fixed 0.1
STS-B	81.4	80.5	71.0	43.6



3. Supervised SimCSE

Contradiction as hard negatives

Given one premise,

- Premise: *There are two dogs running.*

Annotators are required to write hypotheses of

- Entailment: *There are animals outdoors.*
- Contradiction: *The pets are sitting on a couch.*
- Neutral: *The dogs are catching a ball.*

- ☑ premise: 하나의 전제
- ☑ Entailment: 그에 따른 가정이 T
- ☑ neutral: 그에 따른 가정이 T/F 알 수 없음
- ☑ contradiction: 그에 따른 가정이 F

$$-\log \frac{e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau}}{\sum_{j=1}^N \left(e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)/\tau} + e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^-)/\tau} \right)}$$

- ☑ 'contradiction' label을 가진 쌍의 문장을
Hard negative pair 로 사용할때 loss function

Choices of labeled data

Dataset	sample	full
Unsup. SimCSE (1m)	-	82.5
QQP (134k)	81.8	81.8
Flickr30k (318k)	81.5	81.4
ParaNMT (5m)	79.7	78.7
SNLI+MNLI		
entailment (314k)	84.1	84.9
neutral (314k) ⁸	82.6	82.9
contradiction (314k)	77.5	77.6
all (942k)	81.7	81.9
SNLI+MNLI		
entailment + hard neg.	-	86.2
+ ANLI (52k)	-	85.0

QQP4 : Quora question pairs 동일한 질문
Flickr30k : 동일한 이미지에서 나온 손글씨
ParaNMT : 영어 문장과, 타언어 문장을 영어로 번역한 문장 Pair
NLI dataset, SNLI (standford 에서 만든 문장 pair),
MNLI (10개 domain 에서 추출한 문장 pair)

- ☑ SNLI, MNLI 가 annotation quality 가 높은 데이터 셋
- ☑ Entailment , Contradiction label 이 잘 define

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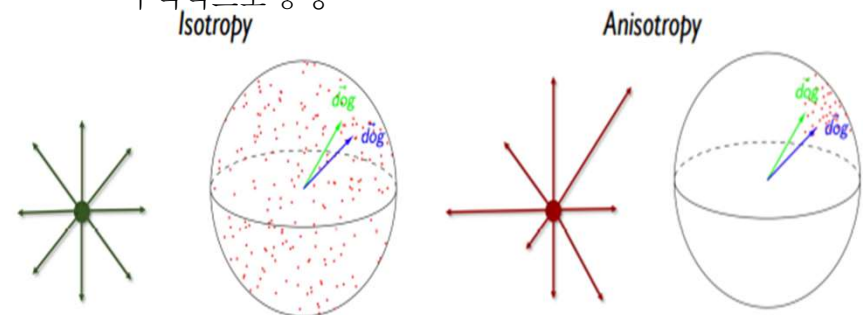
Connection to Anisotropy

$$\mathcal{L}_{\text{contrastive}}(f; \tau, M) \triangleq \mathbb{E}_{\substack{(x, y) \sim p_{\text{pos}} \\ \{x_i^-\}_{i=1}^M \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}}} \left[-\log \frac{e^{f(x)^\top f(y)/\tau}}{e^{f(x)^\top f(y)/\tau} + \sum_i e^{f(x_i^-)^\top f(y)/\tau}} \right] \\ - \frac{1}{\tau} \mathbb{E}_{(x, x^+) \sim p_{\text{pos}}} [f(x)^\top f(x^+)] \\ + \mathbb{E}_{x \sim p_{\text{data}}} \left[\log \mathbb{E}_{x^- \sim p_{\text{data}}} [e^{f(x)^\top f(x^-)/\tau}] \right], \quad (6)$$

$$\mathbb{E}_{x \sim p_{\text{data}}} \left[\log \mathbb{E}_{x^- \sim p_{\text{data}}} [e^{f(x)^\top f(x^-)/\tau}] \right] \\ = \frac{1}{m} \sum_{i=1}^m \log \left(\frac{1}{m} \sum_{j=1}^m e^{\mathbf{h}_i^\top \mathbf{h}_j / \tau} \right) \quad (7) \\ \geq \frac{1}{\tau m^2} \sum_{i=1}^m \sum_{j=1}^m \mathbf{h}_i^\top \mathbf{h}_j.$$

$$\text{Sum}(\mathbf{W}\mathbf{W}^\top) = \sum_{i=1}^m \sum_{j=1}^m \mathbf{h}_i^\top \mathbf{h}_j$$

- ☑ contrastive learning이 anisotropy 문제를 해결할 수 있음이 수식적으로 증명



- ☑ Anisotropy 는 embedding expressiveness 제한

5 Experiments & Analysis

- ✓ Experiments
 - Main result
 - Ablation studies
- ✓ Analysis
 - Align & Uniform

5. Experiments

Main result

Model	STS12	STS13	STS14	STS15	STS16	STS-B	SICK-R	Avg.
<i>Unsupervised models</i>								
GloVe embeddings (avg.) [★]	55.14	70.66	59.73	68.25	63.66	58.02	53.76	61.32
BERT _{base} (first-last avg.)	39.70	59.38	49.67	66.03	66.19	53.87	62.06	56.70
BERT _{base} -flow	58.40	67.10	60.85	75.16	71.22	68.66	64.47	66.55
BERT _{base} -whitening	57.83	66.90	60.90	75.08	71.31	68.24	63.73	66.28
IS-BERT _{base} [♡]	56.77	69.24	61.21	75.23	70.16	69.21	64.25	66.58
CT-BERT _{base}	61.63	76.80	68.47	77.50	76.48	74.31	69.19	72.05
* SimCSE-BERT _{base}	68.40	82.41	74.38	80.91	78.56	76.85	72.23	76.25
RoBERTa _{base} (first-last avg.)	40.88	58.74	49.07	65.63	61.48	58.55	61.63	56.57
RoBERTa _{base} -whitening	46.99	63.24	57.23	71.36	68.99	61.36	62.91	61.73
DeCLUTR-RoBERTa _{base}	52.41	75.19	65.52	77.12	78.63	72.41	68.62	69.99
* SimCSE-RoBERTa _{base}	70.16	81.77	73.24	81.36	80.65	80.22	68.56	76.57
* SimCSE-RoBERTa _{large}	72.86	83.99	75.62	84.77	81.80	81.98	71.26	78.90
<i>Supervised models</i>								
InferSent-GloVe [★]	52.86	66.75	62.15	72.77	66.87	68.03	65.65	65.01
Universal Sentence Encoder [★]	64.49	67.80	64.61	76.83	73.18	74.92	76.69	71.22
SBERT _{base} [★]	70.97	76.53	73.19	79.09	74.30	77.03	72.91	74.89
SBERT _{base} -flow	69.78	77.27	74.35	82.01	77.46	79.12	76.21	76.60
SBERT _{base} -whitening	69.65	77.57	74.66	82.27	78.39	79.52	76.91	77.00
CT-SBERT _{base}	74.84	83.20	78.07	83.84	77.93	81.46	76.42	79.39
* SimCSE-BERT _{base}	75.30	84.67	80.19	85.40	80.82	84.25	80.39	81.57
SRoBERTa _{base} [★]	71.54	72.49	70.80	78.74	73.69	77.77	74.46	74.21
SRoBERTa _{base} -whitening	70.46	77.07	74.46	81.64	76.43	79.49	76.65	76.60
* SimCSE-RoBERTa _{base}	76.53	85.21	80.95	86.03	82.57	85.83	80.50	82.52
* SimCSE-RoBERTa _{large}	77.46	87.27	82.36	86.66	83.93	86.70	81.95	83.76

☑ Contrastive learning 다른 방법론보다 좋은 성능

Ablation studies

Pooler	Unsup.	Sup.
[CLS]		
w/ MLP	81.7	86.2
w/ MLP (train)	82.5	85.8
w/o MLP	80.9	86.2
First-last avg.	81.2	86.1

☑ Pooler method에 영향을 받지 않음

☑ Unsupervised -> 학습때만 MLP + [CLS]
Supervised -> 학습/평가 MLP + [CLS]

Hard negatives

Hard neg	N/A	Contradiction			Contra.+ Neutral
α	-	0.5	1.0	2.0	1.0
STS-B	84.9	86.1	86.2	86.2	85.3

Table 7: STS-B development results with different hard negative policies. “N/A”: no hard negative.

$$-\log \frac{e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau}}{\sum_{j=1}^N \left(e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)/\tau} + \alpha^j e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^-)/\tau} \right)},$$

☑ hard negative 를 contradiction 으로 했을때 성능향상

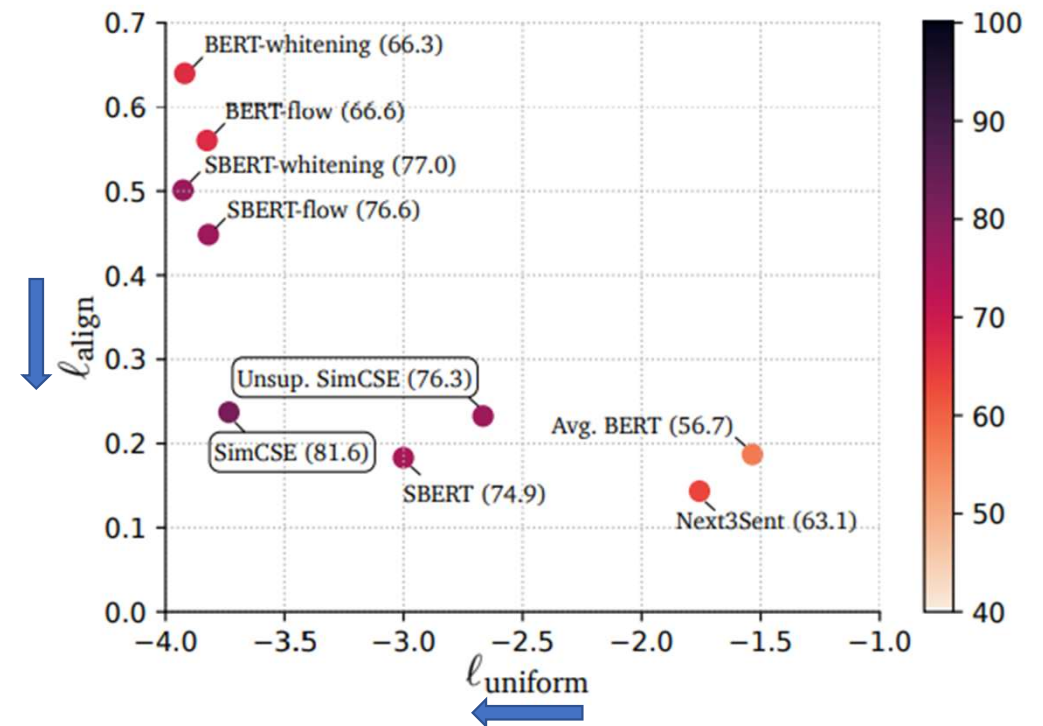
5. Analysis

Qualitative comparison

	SBERT _{base}	Supervised SimCSE-BERT _{base}
Query: A man riding a small boat in a harbor.		
#1	A group of men traveling over the ocean in a small boat.	A man on a moored blue and white boat.
#2	Two men sit on the bow of a colorful boat.	A man is riding in a boat on the water.
#3	A man wearing a life jacket is in a small boat on a lake.	A man in a blue boat on the water.
Query: A dog runs on the green grass near a wooden fence.		
#1	A dog runs on the green grass near a grove of trees.	The dog by the fence is running on the grass.
#2	A brown and white dog runs through the green grass.	Dog running through grass in fenced area.
#3	The dogs run in the green field.	A dog runs on the green grass near a grove of trees.

Table 8: Retrieved top-3 examples by SBERT and supervised SimCSE from Flickr30k (150k sentences).

Alignment & Uniform



☑ SimCSE 가 align, Uniform 측면에서 좋은 성능

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Conclusion

- ✓ Simple contrastive learning framework 제안
 - high quality sentence representation 생성
- ✓ Unsupervised, Supervised contrastive learning 제안
- ✓ Downstream task 와 더불어 성능향상을 Alignment , Uniformity 에서 분석